National University of Computer and Emerging Sciences
Islamabad Operating System Lab Spring 2025

Lab Task Threads

Lab Task 1: Multi-Threaded Exam Grading System (Random Scores)

Each thread represents a teacher grading a student's exam.

Requirements:

- Define a `struct Student` with attributes: `ID`, `score`, and `result`.
- Ask the user to enter the number of students (5 to 10).
- Assign each student a **random score** (between 50-100).
- Each thread processes a student's score and determines **Pass (≥60)** or **Fail (<60)**.
- Use `pthread_join` to ensure all grades are processed before printing "**All exam papers have been graded!**".

Lab Task 2: Multi-Threaded Banking Transaction System

Each thread represents a bank officer handling transactions for **two customers** at the same time.

Requirements:

- Define a `struct Transaction` with attributes: `customer1Name`, `customer2Name`, `amount1`, `amount2`,

`transactionType1`, `transactionType2`.

- Ask the user to enter the number of **customer pairs** (e.g., 3 to 5 pairs, meaning 6 to 10 customers).
- Each **thread processes two customers**:
    - **Customer 1** performs a **deposit or withdrawal** of `amount1`.
    - **Customer 2** performs a **deposit or withdrawal** of `amount2`.
- Print transaction details for both customers handled by each thread.
- Use `pthread_join` to ensure all transactions are processed before printing "**All banking transactions have been completed!**".

Lab Task 3: Multi-Threaded Customer Feedback Collection

Each thread represents a customer submitting feedback about a service.

Requirements:

- Define a `struct Feedback` with attributes: `customerName`, `serviceType`, `rating` (1-5).
- Ask the user to enter the number of customers (5 to 10).
- Each customer provides their name, selects a service (e.g., "Delivery", "Support", "Billing"), and enters a

rating (manually).

- Each thread processes the feedback and stores it for review.

- Use `pthread_join` to ensure all feedback is collected before printing "**All customer feedback has been recorded!**".

THE END